

Systeme de Versionnage de Specifications - Technical Specification & Architecture Document

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1. Executive Summary & Architecture Overview

1.1 Executive Brief

This project implements a technical versioning system designed to validate the upgrade pipeline from v1.0 to v2.0. It utilizes a FastAPI-driven data flow to detect source modifications via SHA-256 hash validation and persists version state updates within a PostgreSQL database, culminating in the generation of optimized PDF specifications.

1.2 Maturity Assessment

The specification is currently in a light test state and requires REFINEMENT. While the core functional flow is mapped, there are critical structural gaps including the absence of a defined project scope and a lack of quantified non-functional requirements (SLAs) for the rapid rendering requirement, which prevents a full production-ready assessment.

1.3 Technical Stack

- FastAPI
- PostgreSQL

1.4 Architectural Constraints

- Version incrementation strictly tied to source file modification detection.
- Data persistence limited to the `doc\_versions` table.
- Zero-latency requirement for enrichment agents during document generation.
- Trigger mechanism based on SHA-256 validation hash.

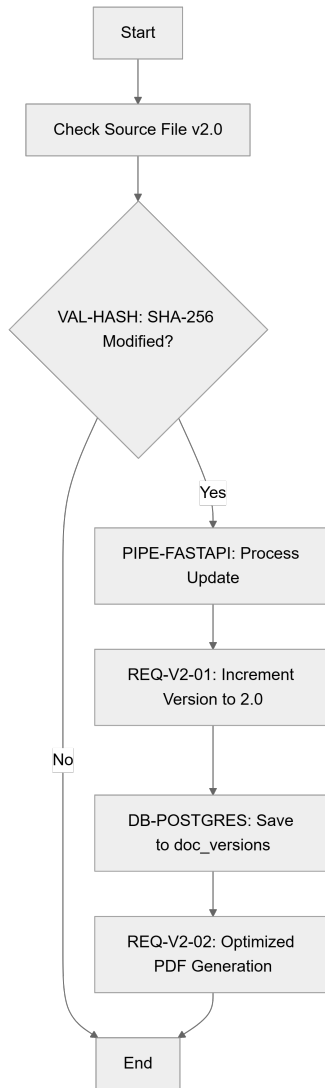
1.5 Critical Dependencies

- SHA-256 hash verification for version trigger.
- PostgreSQL `doc\_versions` table for state recording.
- FastAPI pipeline for data orchestration.
- Correct release of BDD locks prior to versioning cycle.

2. Architecture Workflows & Visual Diagrams

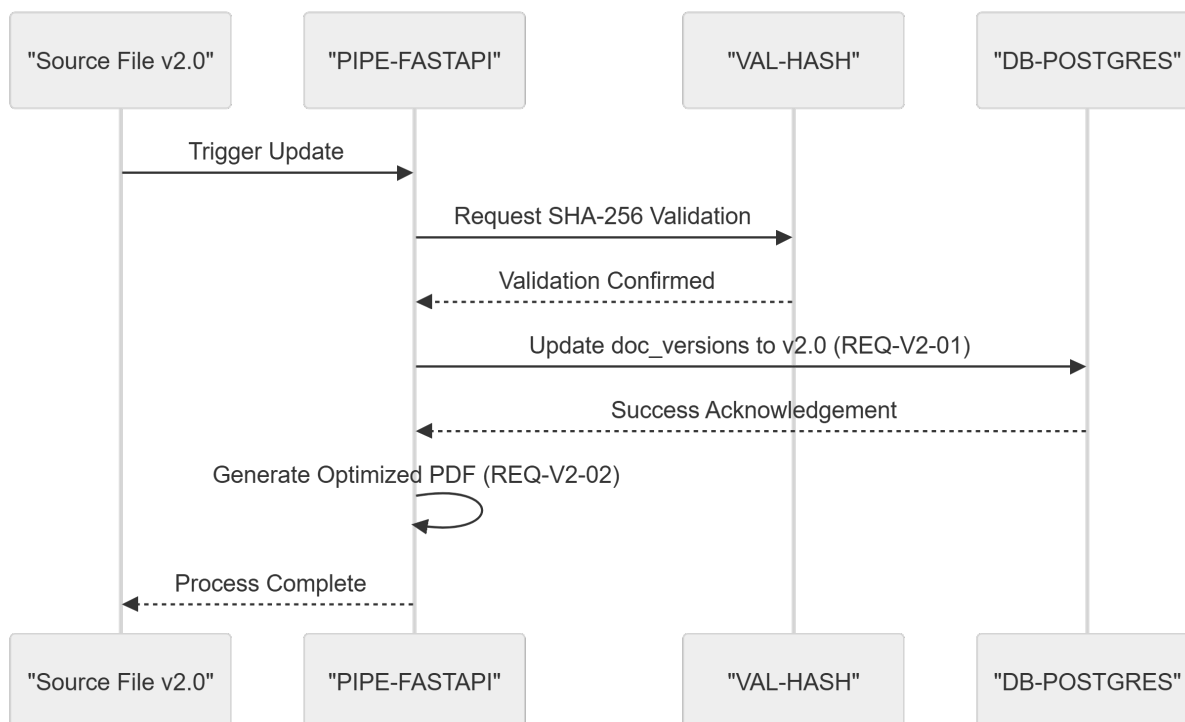
2.1 Version Update Workflow

Models the technical process of detecting file changes and updating the version in the database, incorporating the SHA-256 validation logic.



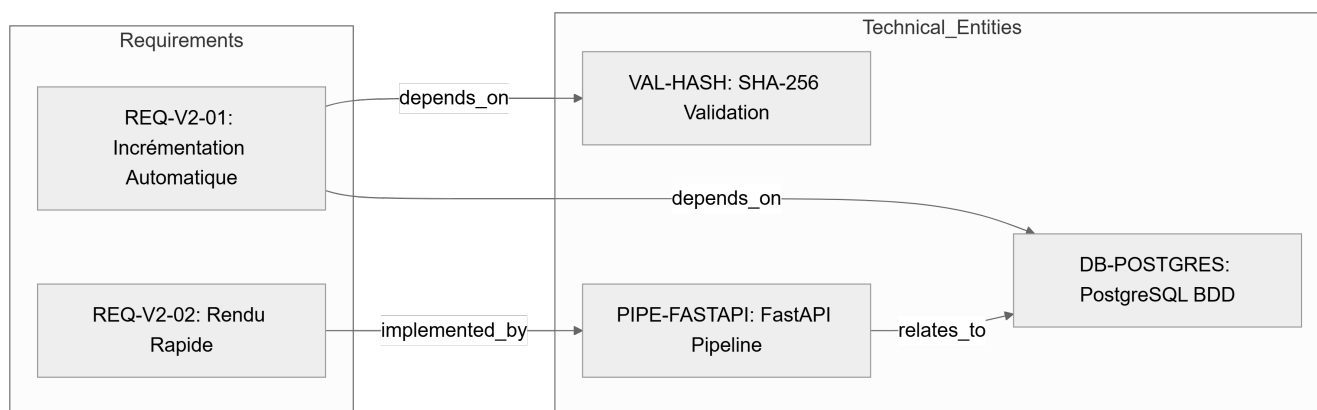
2.2 System Interaction Sequence

Sequence of interactions between the source file, the FastAPI pipeline, and the PostgreSQL database for the versioning process.



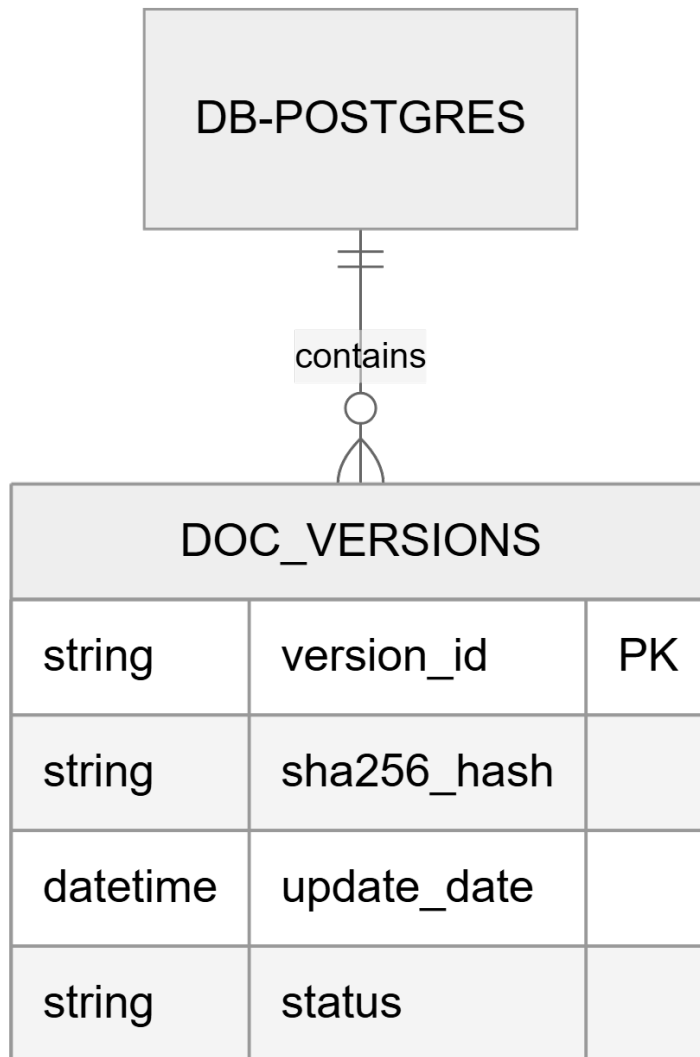
2.3 Requirements Traceability Map

Maps functional requirements to the technical entities they depend on for implementation.



2.4 Data Model Entity Relationship

Conceptual data model for the versioning system focusing on the `doc\_versions` table.



3. Detailed Technical Specifications & Business Rules

3.1 Requirements Traceability

Identifiant	Requirement Description	Technical Entity / Dependency	Feature
REQ-V2-01	Le système doit détecter la modification du fichier source et enregistrer la version 2.0 dans la table doc_versions.	DB-POSTGRES, VAL-HASH	Incrémement Automatique

REQ-V2-02	Génération optimisée du document sans latence sur les agents d'enrichissement.	PIPE-FASTAPI	Rendu Rapide
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3.2 Security Rules

- ****Integrity Validation****: Use of SHA-256 cryptographic hashing (`VAL-HASH`) to ensure that version increments are only triggered by actual content modifications.

3.3 Data Models

Entity ID	Description	Context
DB-POSTGRES	Base de données PostgreSQL contenant la table doc_versions	Persistence Layer
PIPE-FASTAPI	Pipeline FastAPI responsable du flux de données	Orchestration Layer
VAL-HASH	Vérification de la modification SHA-256 pour déclencher la mise à jour	Validation Layer

4. Project Governance & Structural Gaps

4.1 Structural Gaps

Missing Section	Priority	Remediation Advice
Non-Functional Requirements	MEDIUM	Préciser les SLAs de performance pour le 'Rendu Rapide' mentionné dans REQ-V2-02.
Scope & Out-of-Scope	HIGH	Définir les limites du test de versionnage pour éviter tout débordement sur la production.
Open Questions & Uncertainties	LOW	Lister les points de blocage potentiels lors de la réinitialisation des verrous BDD.

4.2 Remediation & Workflow

The project is currently in a "Light Test" phase. To move toward a production-ready specification, the identified gaps in Section 4.1 must be addressed by defining quantitative performance metrics and explicit boundary conditions for the versioning pipeline.

5. Technical & Domain Glossary (Terminology Reference)

Term	Category	Context Anchor	Project Definition
2.0	BUSINESS_DOMAIN	REQ-V2-01	The specific numerical target state for the current update cycle to be persisted in the tracking table.
BDD	TECHNICAL_STACK	■ Spécification Technique du Système — Version 2.0 (Test Léger)	The relational persistence layer where lock resets are performed to allow new record insertions.
Cryptographic Hashing	TECHNICAL_STACK	VAL-HASH	The algorithmic process used to generate a unique fingerprint of the source file to detect modifications.
PDF	TECHNICAL_STACK	1. Présentation Générale	The final read-only document format generated by the enrichment agents for archival purposes.
REQ	BUSINESS_DOMAIN	2. Exigences Fonctionnelles	The standardized prefix used to identify a formal functional constraint within the system topology.
SHA	TECHNICAL_STACK	VAL-HASH	The specific 256-bit secure digest standard employed to trigger the update pipeline.
Validation Hash	TECHNICAL_STACK	VAL-HASH	The process of comparing current and previous file signatures to determine if the update sequence must execute.
version 2.0	BUSINESS_DOMAIN	1. Présentation Générale	The designated milestone representing the transition from the initial release to the current validated state.